Cola and Calcium

MATH 213

Introduction

A sample of 16 healthy women aged 18- 40 were randomly assigned to drink 24 ounces of either diet cola or water. Their urine was collected for three hours after ingestion of the beverage and calcium excretion (in mg.) was measured. The researchers were investigating whether diet cola leaches calcium out of the system, which would increase the amount of calcium in the urine for diet cola drinkers. You can think of the “cola” group as the treatment group and the “water” group as the control group.

Conclusion (Malec wording)

Assuming there is no difference in urine calcium between soda drinkers and water drinkers, our difference in means of 6.875 mg is extremely unusual. Out of 2000 simulated experiments, only 8 had results as extreme as this one, resulting in a p-value of 0.004. Therefore, the probability of this result occurring due to chance or random varation is very small, which means that we have evidence to reject the null hypothesis. In other words, this experiment is evidence that female cola drinkers may excrete more calcium than the water drinkers.

Group 1

1. Who or what are the cases for this study?

    The cases would be the 16 women

2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

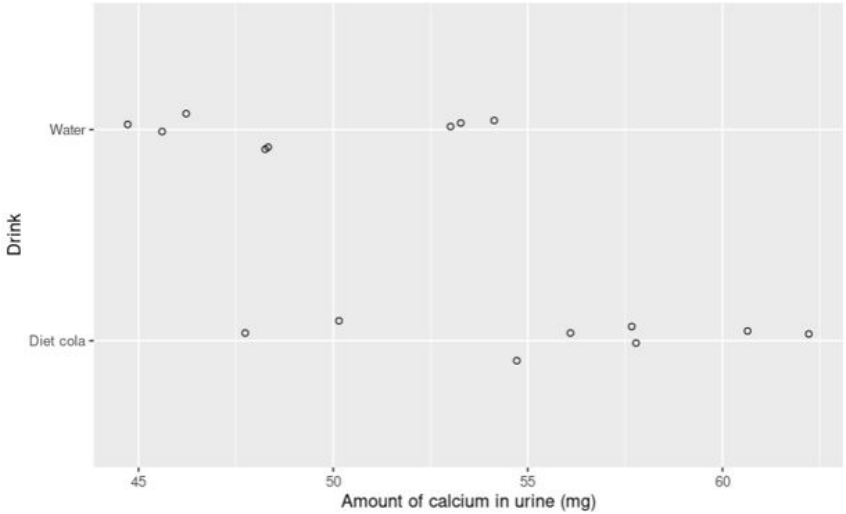
    Explanatory: Drink (categorical)

    Response: Concentration of calcium in urine (mg) (numerical)

3. Was this an experiment or an observational study?

    Experiment

Here is a graph of the experimental results:



And here is some more information:

And here is some more information:

```{r}
cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more calcium in their urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

The beverage assigned had no effect on the amount of calcium in the participant’s urine

$\mu_{dietCola} - \mu_{water} = 0$

7. Write an alternative hypothesis for this situation:

The beverage assigned did have an effect: the cola caused participant’s urine to have a higher concentration of calcium

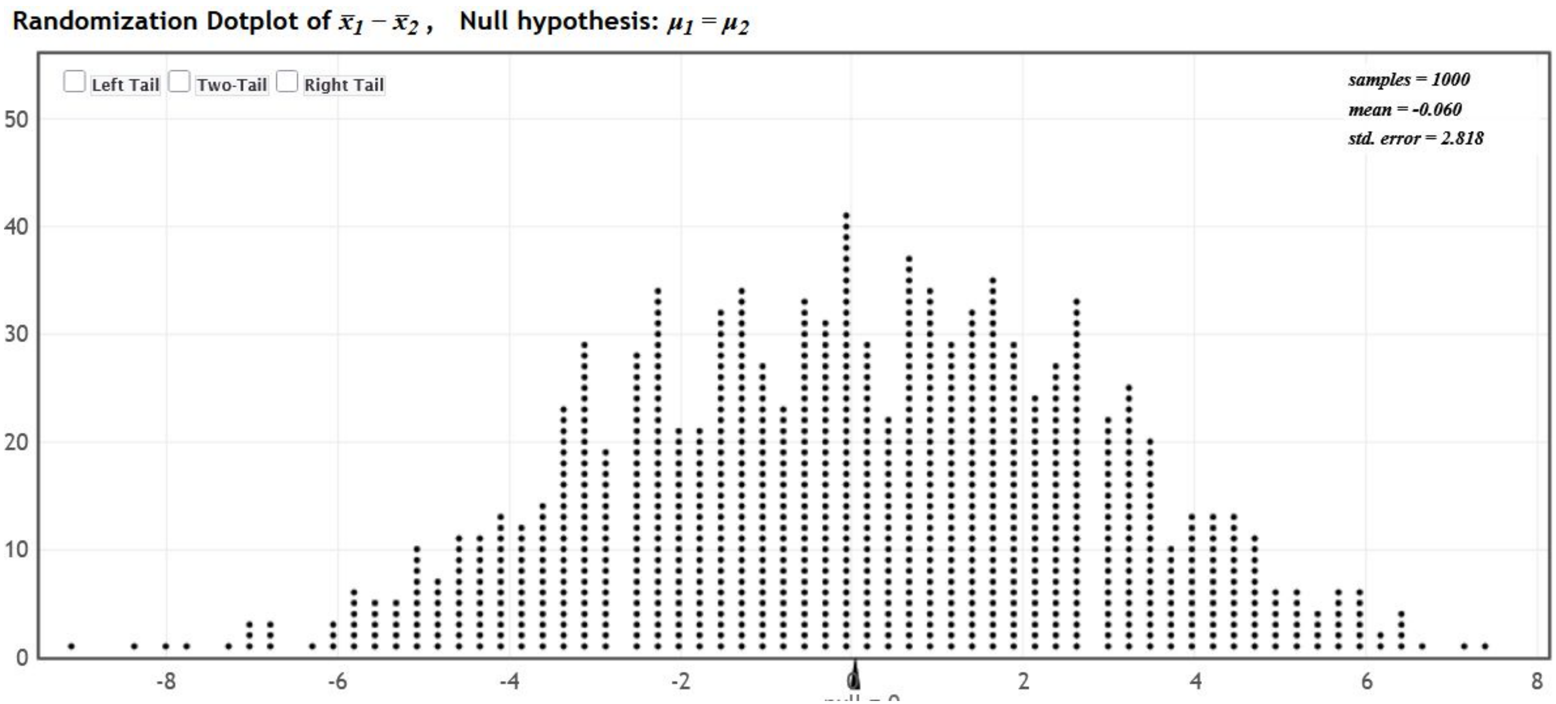
$\mu_{dietCola} - \mu_{water} > 0$

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

Each dot is a sample of the experiment

10. How many dots are in your dotplot?

1000

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

6.875 does seem unusual in comparison

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

we have a a p-value of 0.002.

P = 2 / 1000

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

We have found that our result of 6.875 is very unusual, with a found p-value of 0.002 from our simulations. Therefore, we will reject the null hypothesis, as the probability of the result occurring due to random variation is extremely small. Because of this, we may have evidence that the diet cola drinkers may have more calcium in their urine sample than water drinkers.

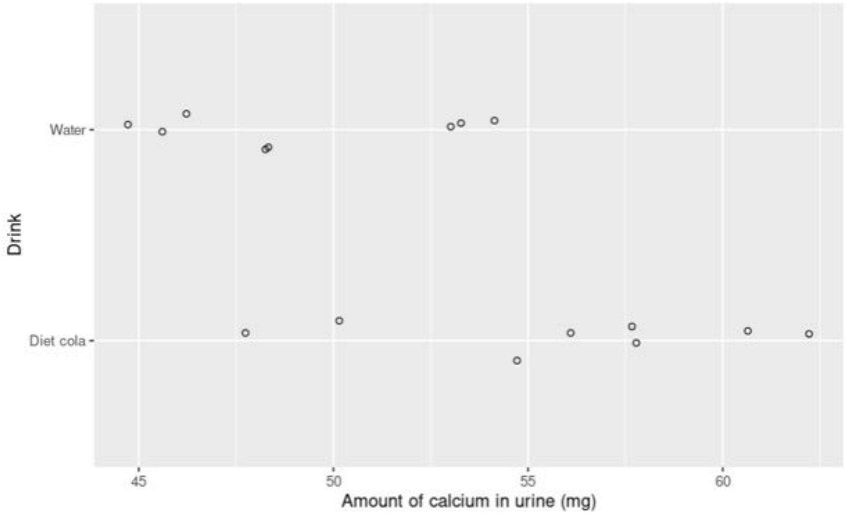
Introduction

A sample of 16 healthy women aged 18- 40 were randomly assigned to drink 24 ounces of either **diet cola or water**. Their urine was collected for three hours after ingestion of the beverage and **calcium excretion (in mg.)** was measured. The researchers were investigating whether diet cola leaches calcium out of the system, which would increase the amount of calcium in the urine for diet cola drinkers. **You can think of the “cola” group as the treatment group and the “water” group as the control group.**

Group 2

1. Who or what are the cases for this study?
    - 16 healthy women aged 18- 40
2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.
    - Explanatory: Treatment group (diet cola or water) – Categorical
    - Response: Calcium excretion in mg – Numerical
3. Was this an experiment or an observational study?
    - Experiment

Here is a graph of the experimental results:



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cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

- Diet cola has no effect on calcium excretion
    - H0: $\mu_{cola} - \mu_{water} = 0$

7. Write an alternative hypothesis for this situation:

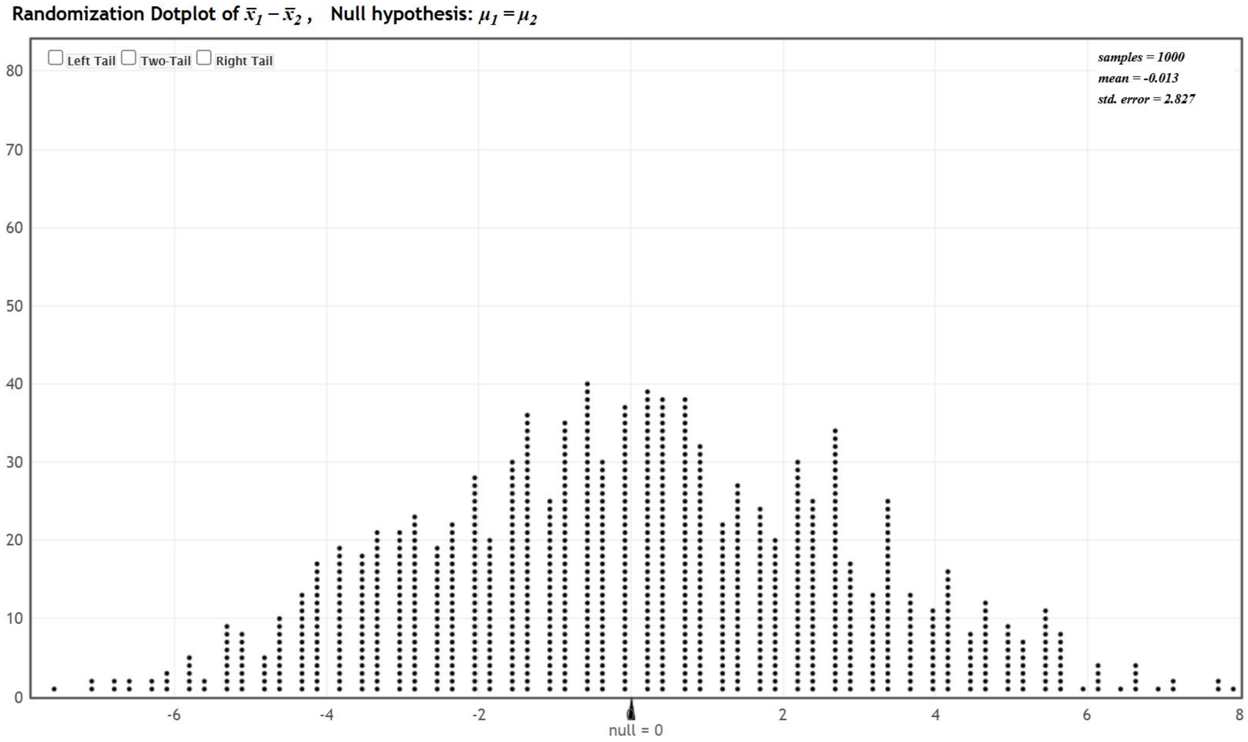
- Diet cola increases calcium excretion
    - Ha: $\mu_{cola} - \mu_{water} > 0$

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

Each dot represents the difference in mean calcium excretion for a particular simulation under the null hypothesis (the assumption that the cola has no effect on calcium excretion).

10. How many dots are in your dotplot?

1000 dots.

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

6.875 seems to be unusual, since there are not many points that are equal to or higher than that value on the dotplot.

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

The p-value we calculated is 0.006.

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

The p-value we calculated is 0.006. This means that the probability of obtaining a result as or more extreme than the experimental results under the assumption of the null hypothesis is 0.6 %.

Group 3

1. Who or what are the cases for this study?

   16 healthy women

2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

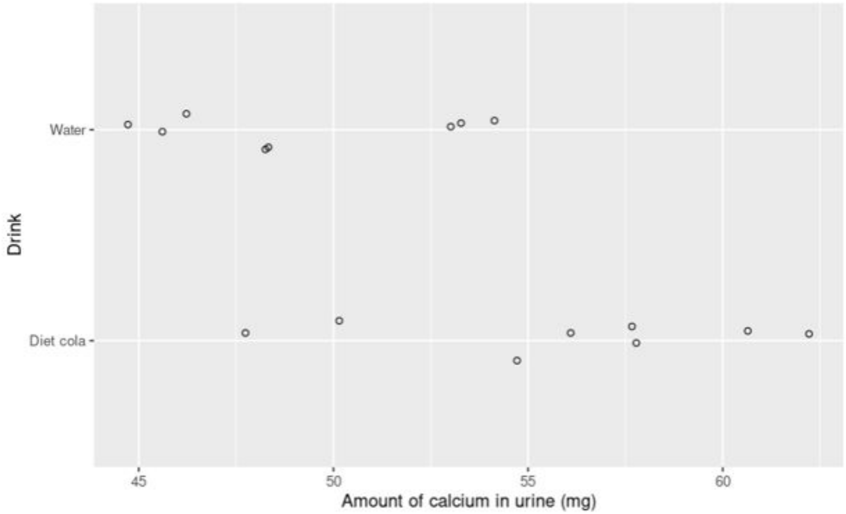
   Explanatory: Type of drink - categorical

   Response: Calcium in urine - numerical

3. Was this an experiment or an observational study?

   Experiment

Here is a graph of the experimental results:



And here is some more information:

And here is some more information:

```{r}
cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

Drinking water and drinking coke have the same effect on calcium concentration in the urine in healthy women.

7. Write an alternative hypothesis for this situation:

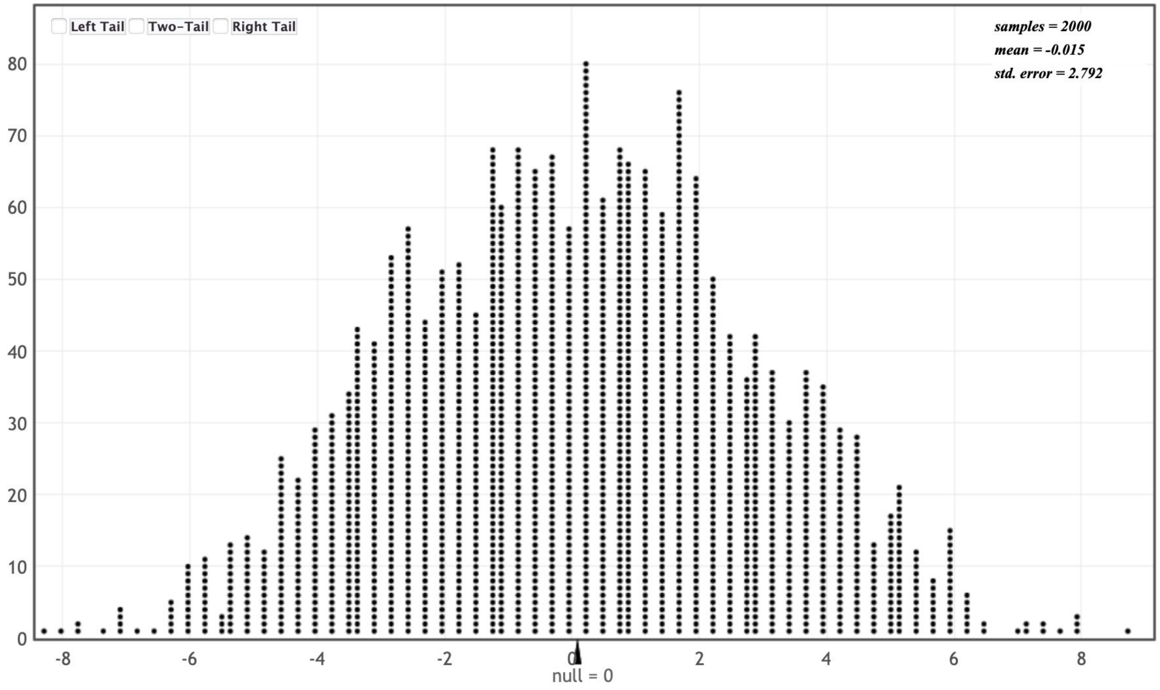
Drinking water and drinking coke do not have the same effect on calcium concentration in the urine in healthy women.

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

The difference in the average calcium secretion between healthy women who drank diet coke vs drank water for one simulation, under the assumption that the null hypothesis is true.

10. How many dots are in your dotplot?

2000 dots.

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

Yes because the p value for the experimental data value is .005, which under the assumption that the null is true would be an extremely unlikely result.

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

P = 0.005

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

We would reject the null hypothesis in this case, as the recorded p value is .005, which is far below the p=.05 threshold in order to reject the null.

Group 4

1. Who or what are the cases for this study?

The 16 healthy women aged 18-40.

2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

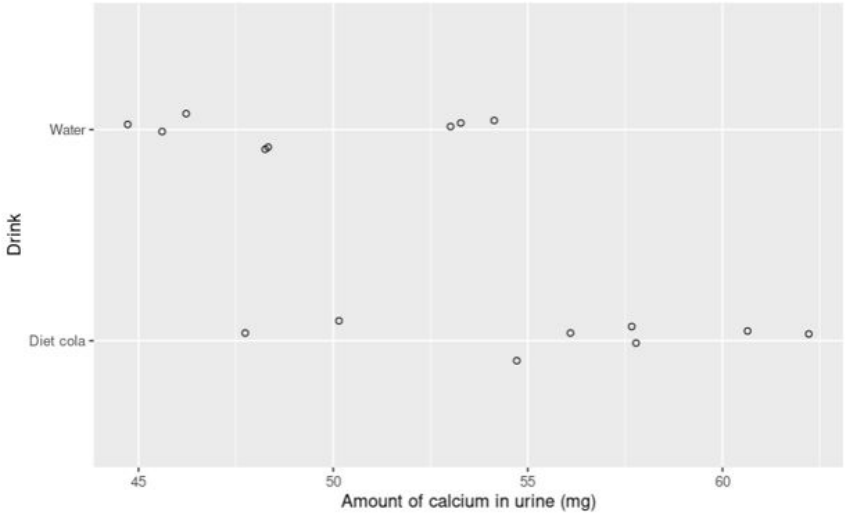
Explanatory: Cola or water (Categorical)

Response: Calcium excretion (mg)

3. Was this an experiment or an observational study?

Experiment

Here is a graph of the experimental results:



And here is some more information:

And here is some more information:

```{r}
cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

Cola has no effect on calcium excretion

$\mu\_treatment - \mu\_control = 0$

7. Write an alternative hypothesis for this situation:

Cola has an effect on calcium excretion

$\mu\_treatment - \mu\_control > 0$

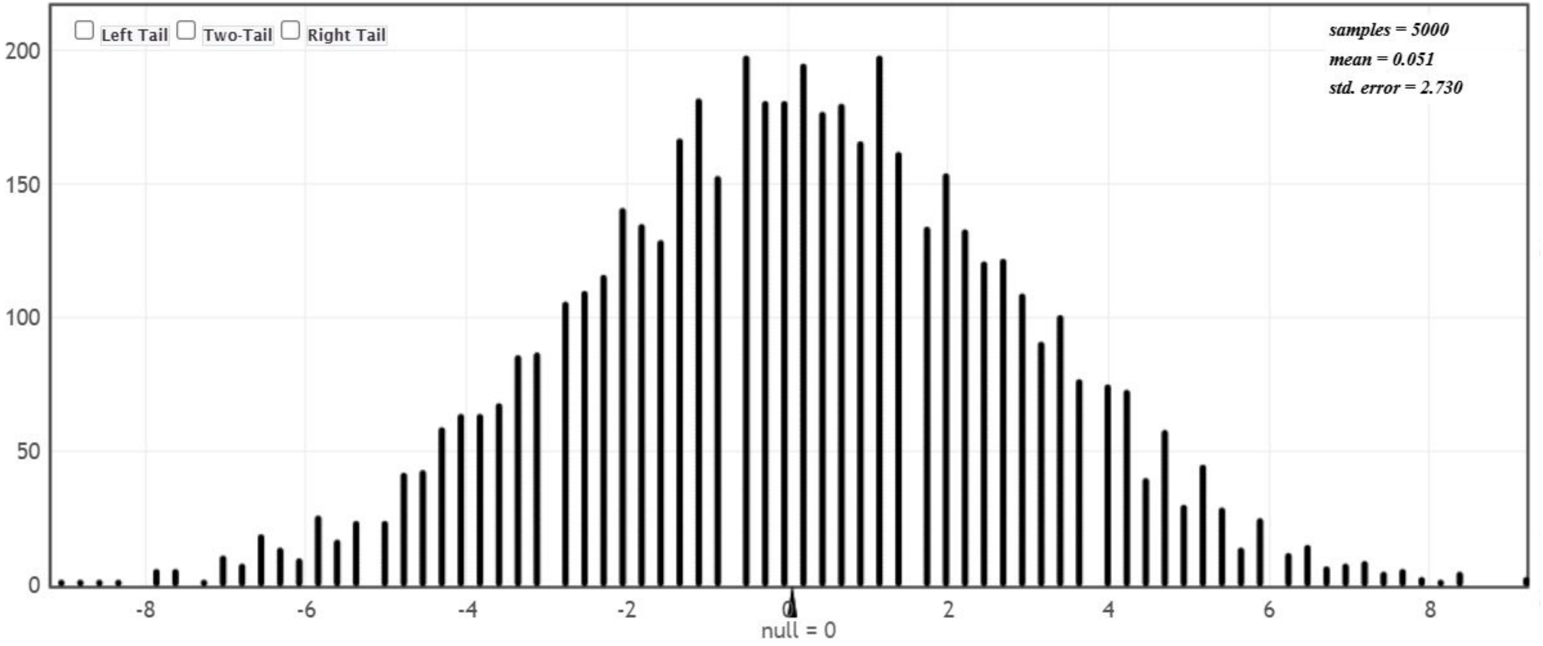
You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.

Randomization Dotplot of $\bar{x}_1 - \bar{x}_2$, Null hypothesis: $\mu_1 = \mu_2$



9. What does each dot in the dotplot represent? Be specific and speak in context.

Each dot represents the difference in the means for one simulated experiment assuming the null is true.

10. How many dots are in your dotplot?

5,000

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

It doesn’t seem too usual

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

0.0066

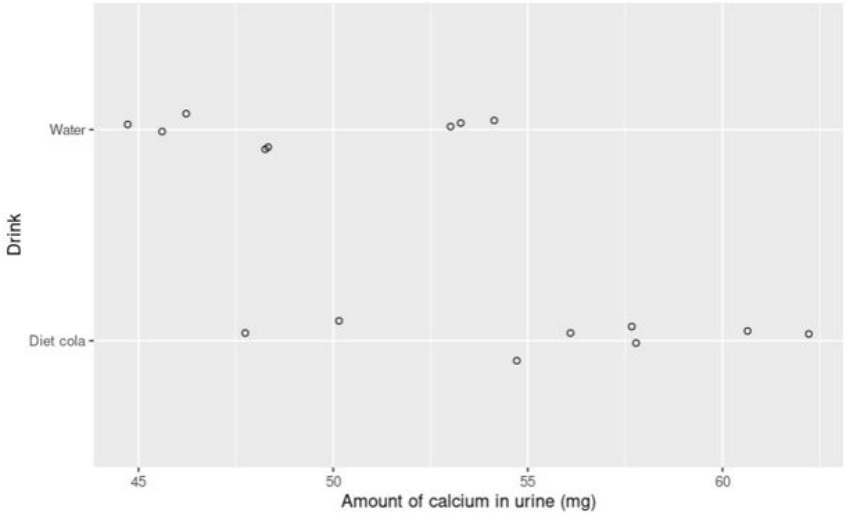
13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

By running a simulation of 5,000 experiments under the null hypothesis, we calculated a p-value for this experiment of 0.0066. Since the p-value is so low at 0.0066, it is unlikely that we would have seen this exact result under the null hypothesis, and so we can reject the null hypothesis. Thus, we can reject the idea that cola has no effect on calcium excretion.

Group 5

1. Who or what are the cases for this study?
2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.
3. Was this an experiment or an observational study?

Here is a graph of the experimental results:



And here is some more information:

And here is some more information:

```
```{r}
cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

7. Write an alternative hypothesis for this situation:

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.

9. What does each dot in the dotplot represent? Be specific and speak in context.

10. How many dots are in your dotplot?

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

Group 6

A sample of 16 healthy women aged 18- 40 were randomly assigned to drink 24 ounces of either diet cola or water. Their urine was collected for three hours after ingestion of the beverage and calcium excretion (in mg.) was measured. The researchers were investigating whether diet cola leaches calcium out of the system, which would increase the amount of calcium in the urine for diet cola drinkers. You can think of the “cola” group as the treatment group and the “water” group as the control group.

1. **Who or what are the cases for this study?**

The 16 women aged 18-40 that were randomly selected.

2. **What is the explanatory variable? The response variable? Classify each as categorical or numerical.**

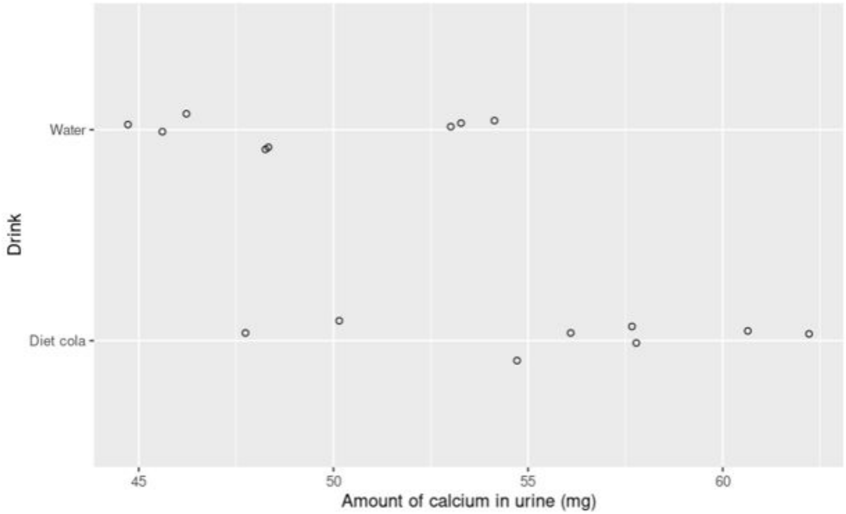
The explanatory variable is the drink they were assigned. (categorical)

The response variable is their urine calcium excretion in mg. (numerical)

3. **Was this an experiment or an observational study?**

This is an experiment because the researchers randomly assigned the participants their drinks which is the explanatory variable.

Here is a graph of the experimental results:



And here is some more information:

And here is some more information:

```
```{r}
cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

Cola has no impact on urine calcium excretion.

7. Write an alternative hypothesis for this situation:

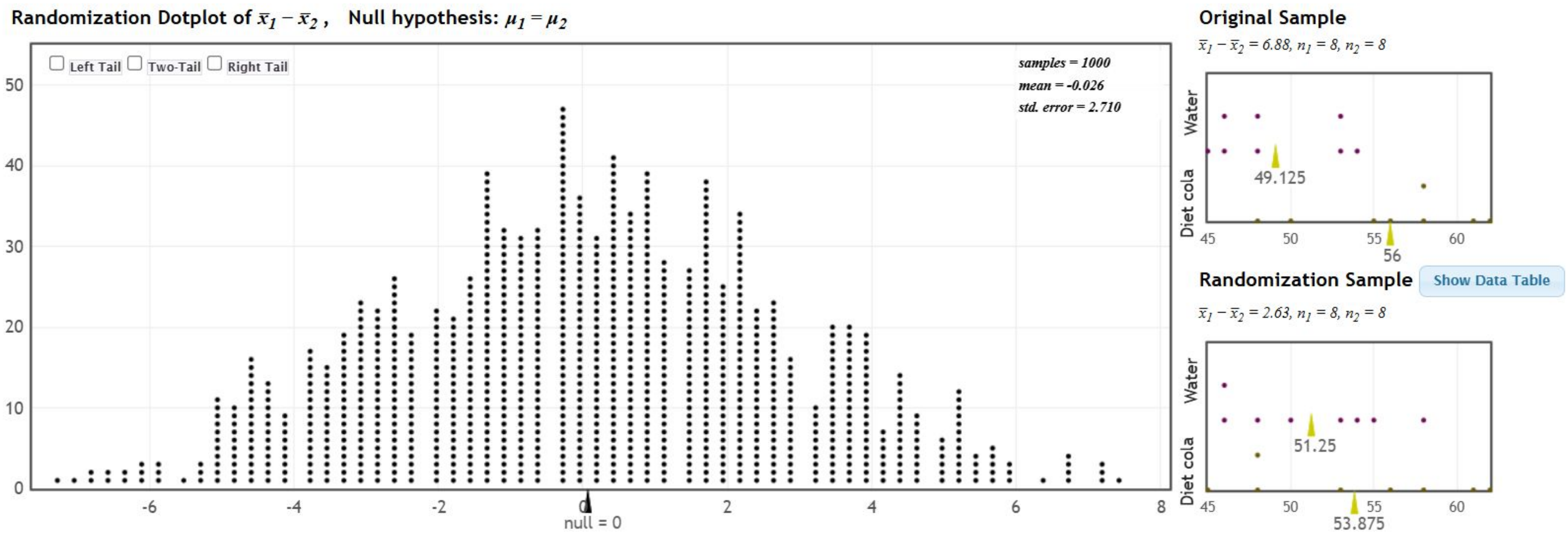
Cola has caused the calcium to be leached from their system and excreted through their urine.

> ***You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.***

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

The difference of means between the simulated control group and sample group for each simulation under the assumption that the null hypothesis is true.

10. How many dots are in your dotplot?

There are 1000 dots in the dotplot.

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

Yes, because the odds of randomly getting a value similar to 6.875mg is roughly only a 0.4% chance of achieving that result.

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

There are 4 values out of a sample size of 1000. This gives a p-value of 0.0040.

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

The randomization test has told us that we cannot accept the null hypothesis. After finishing the randomization test, we found that there was only a 0.4% chance that we would have gotten similar results assuming the null hypothesis is true. Because our actual result is so profoundly unusual, we should reject the null hypothesis. Further testing would be required to confirm our alternate hypothesis.

Group 7

1. Who or what are the cases for this study?

A sample of 16 healthy women aged 18- 40

2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

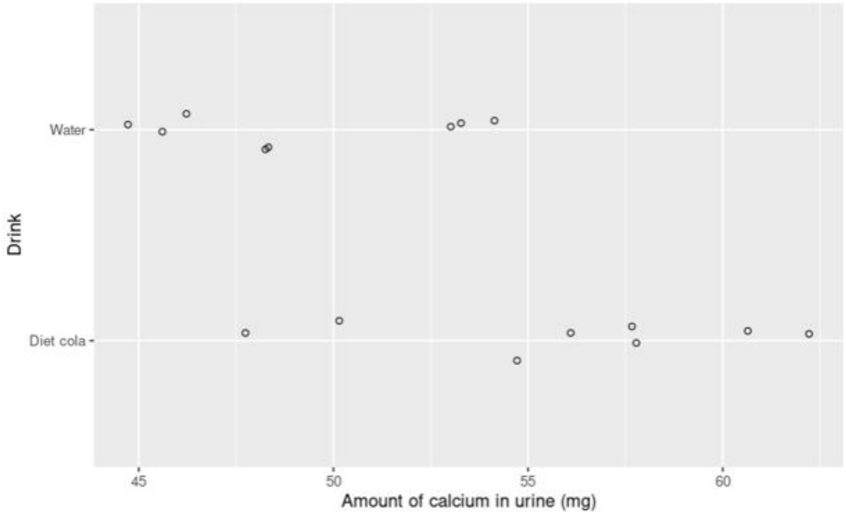
   Explanatory variable is whether or not cola was consumed

   Response was levels of calcium in urine

3. Was this an experiment or an observational study?

   This is an experiment because researchers are controlling the group assignments

Here is a graph of the experimental results:



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```{r}
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  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

    Drinking of cola did not affect levels mean (x-bar) of calcium in urine

    Mu = 49 (mean level water)

7. Write an alternative hypothesis for this situation:

    Drinking cola increases calcium in urine

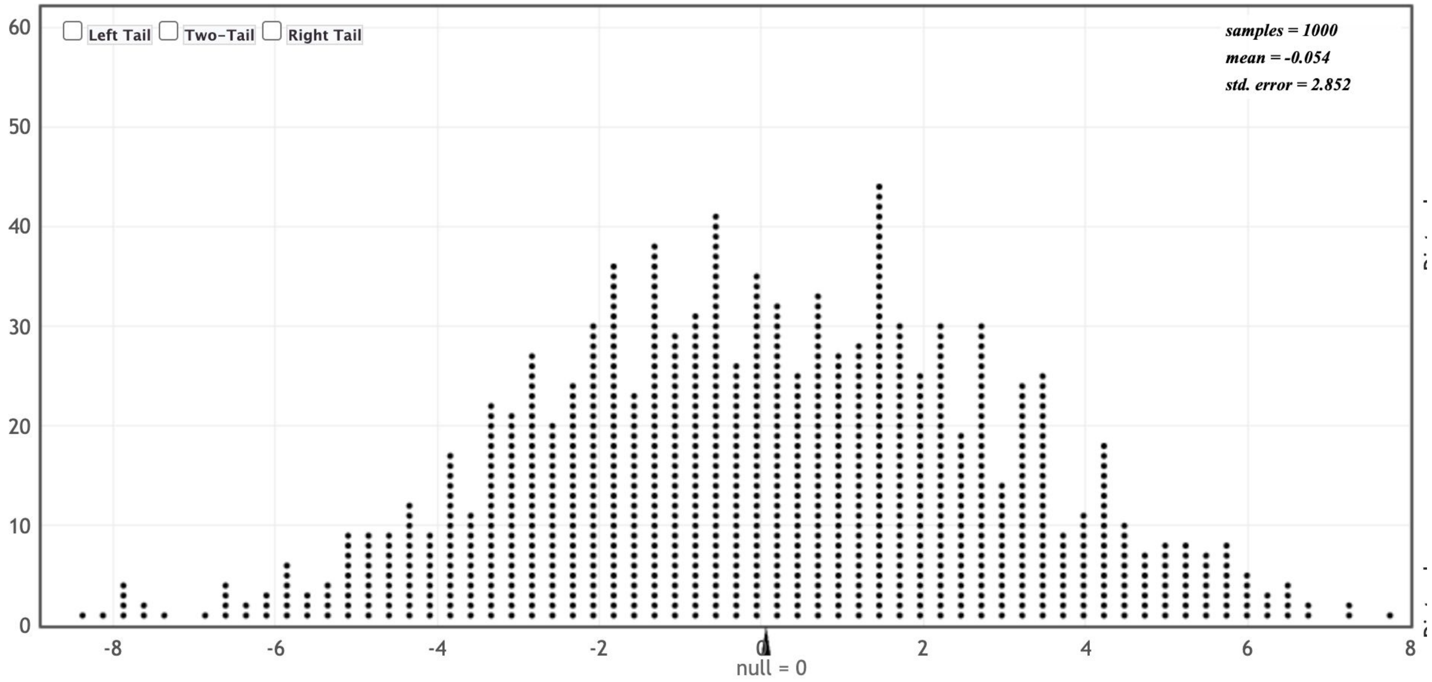
    Mu > 49(mean water)

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

Each dot represents an experiment under the assumption that the null hypothesis is true

10. How many dots are in your dotplot?

1000

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

If calcium levels are above 6.875mg, then there is no correlation between soda drinking and calcium levels

0.003

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

A p_value of 0.003 indicates this is unusual for 6.875mg as only 8 experiments out of 2000 simulations had results as extreme as this. Therefore the odds of this happening to chance is very low.

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

We

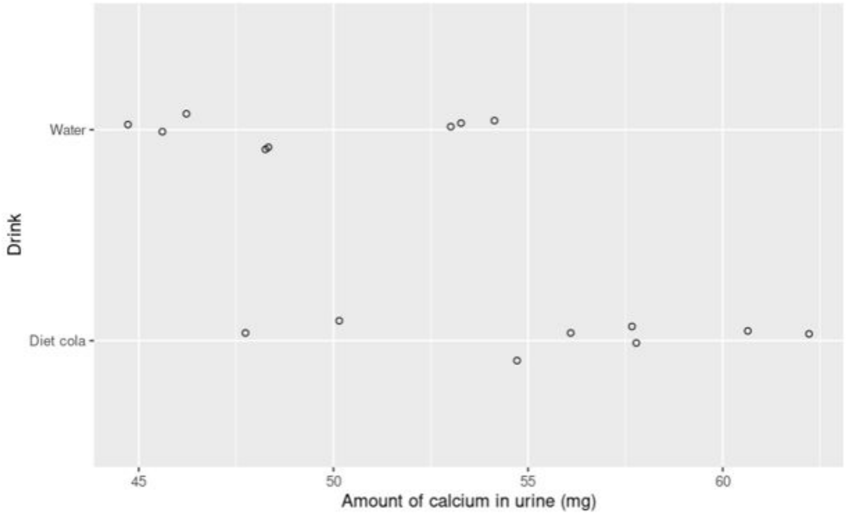
Group 8

1. Who or what are the cases for this study?

2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

3. Was this an experiment or an observational study?

Here is a graph of the experimental results:



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And here is some more information:

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  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
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A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

7. Write an alternative hypothesis for this situation:

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.

9. What does each dot in the dotplot represent? Be specific and speak in context.

10. How many dots are in your dotplot?

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

Group 9

1. Who or what are the cases for this study?

   The 16 women

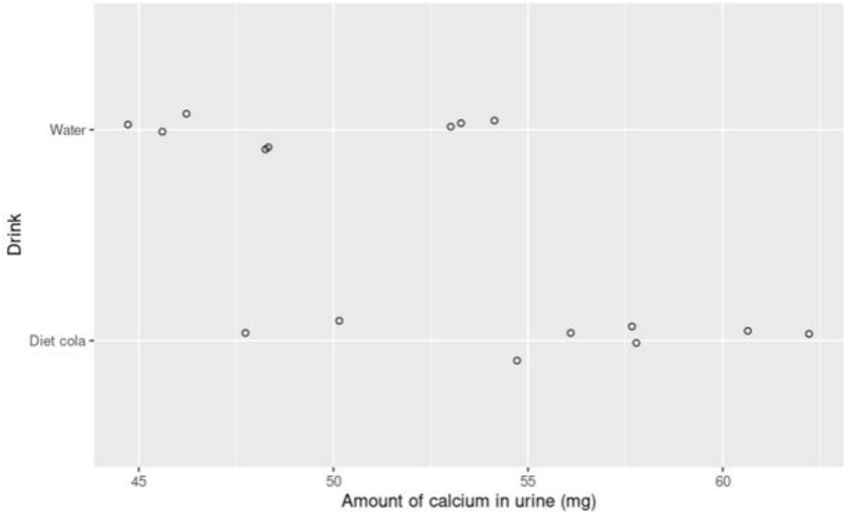
2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

   The groups/What they drink is explanatory, and it is categorical. The response variable is calcium excretion and that is numerical

3. Was this an experiment or an observational study?

   Experiment because they were given groups.

Here is a graph of the experimental results:



And here is some more information:

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```
```{r}
cola_calcium %>%
  group_by(Drink)%>%
  summarize(mean_calcium = mean(Calcium))
```
```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation:

The different drinks assigned to each of the women had no effect on calcium excretion

7. Write an alternative hypothesis for this situation:

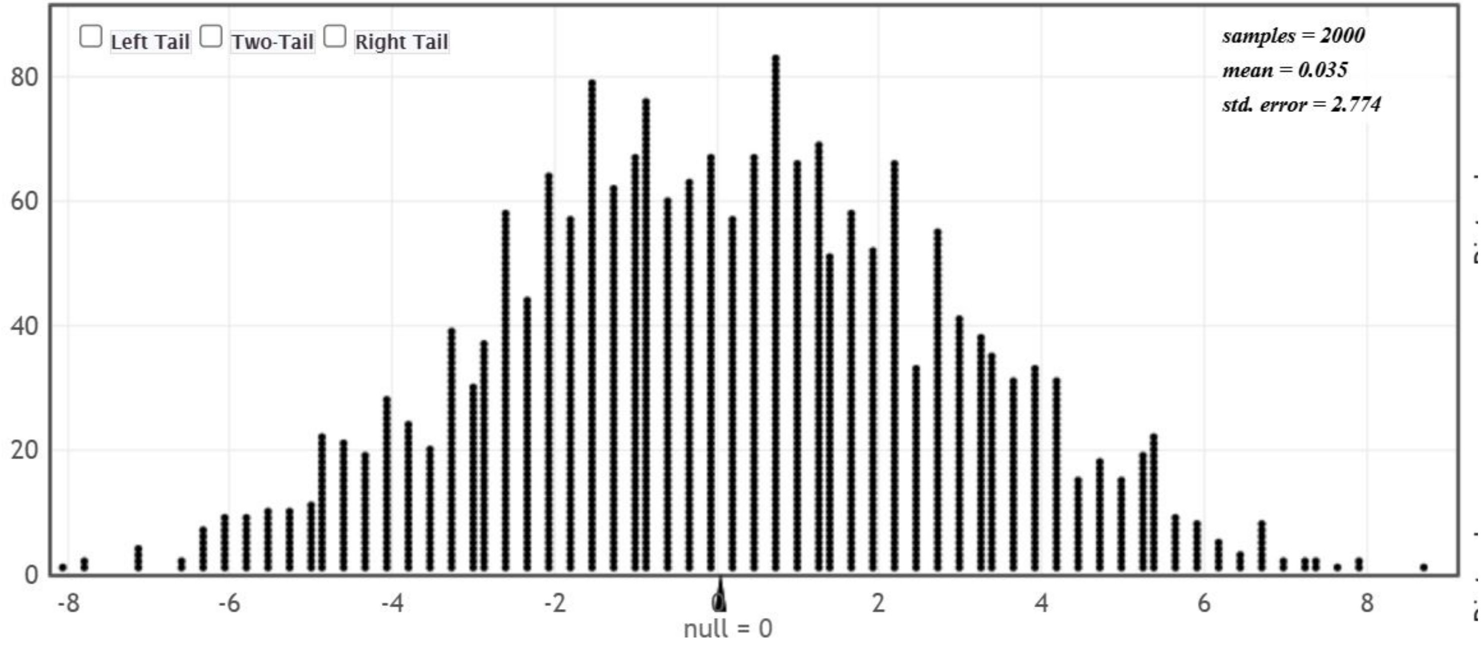
You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

The drink assigned to each of the women did have an effect on calcium excretion: the women who drank cola did excret more calcium

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

The dots represent the difference in means between the cola group and the water group per sample assuming the null hypothesis is true

10. How many dots are in your dotplot?

2000

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

It does seem unusual because that point in the dot plot is at the tail end of the bell curve, meaning it’s unlikely

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots.

10/2000=.005

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

If we were to retain the null hypothesis, the chance of obtaining a result greater than or equal to 6.875 out of a random sample of 2000 is .5%. This means that it is far more likely that this experiment result rejects the null hypothesis

Group 10

1. Who or what are the cases for this study? The cases are the 16 woman

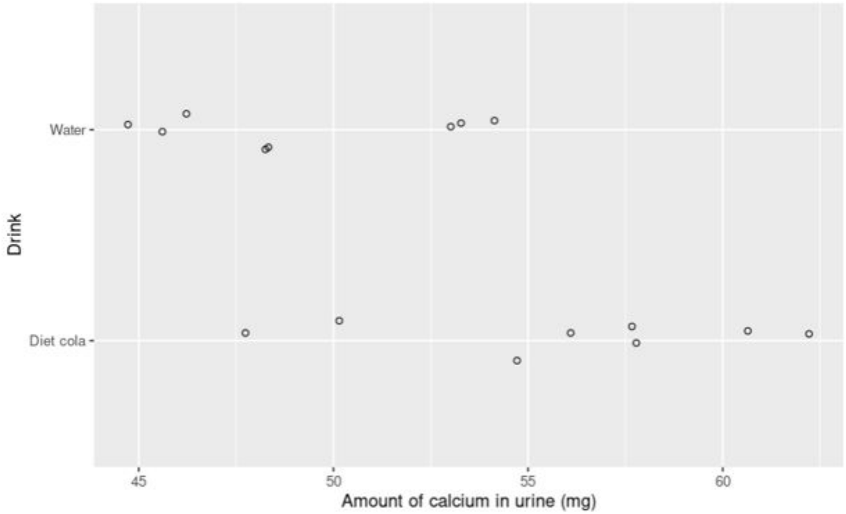
2. What is the explanatory variable? The response variable? Classify each as categorical or numerical.

   The explanatory is which substance they drank(categorical) and the response is the amount of calcium in the urine

3. Was this an experiment or an observational study?

   This would be a experiment because they are being put in groups, which creates a control and a treatment group.

Here is a graph of the experimental results:



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```

A tibble: 2 × 2

Drink <chr>	mean_calcium <dbl>
Diet cola	56.000
Water	49.125

2 rows

From the experimental data we see that in the study, those who drank diet cola tended to excrete more urine than those in the control group.

But I am skeptical! What if the difference we see between the groups occurred by accident, due to the process of random assignment and not because drinking diet cola had an effect on calcium excretion? We will perform a hypothesis test to check if we think this could have happened.

6. Write a null hypothesis for this situation: The calcium excretion in the two groups is going to be the same between the difference in the drinks. The difference in the mean is coming from the random assignment of woman to diet cola and the water groups. Ho: Mc-Mw=0

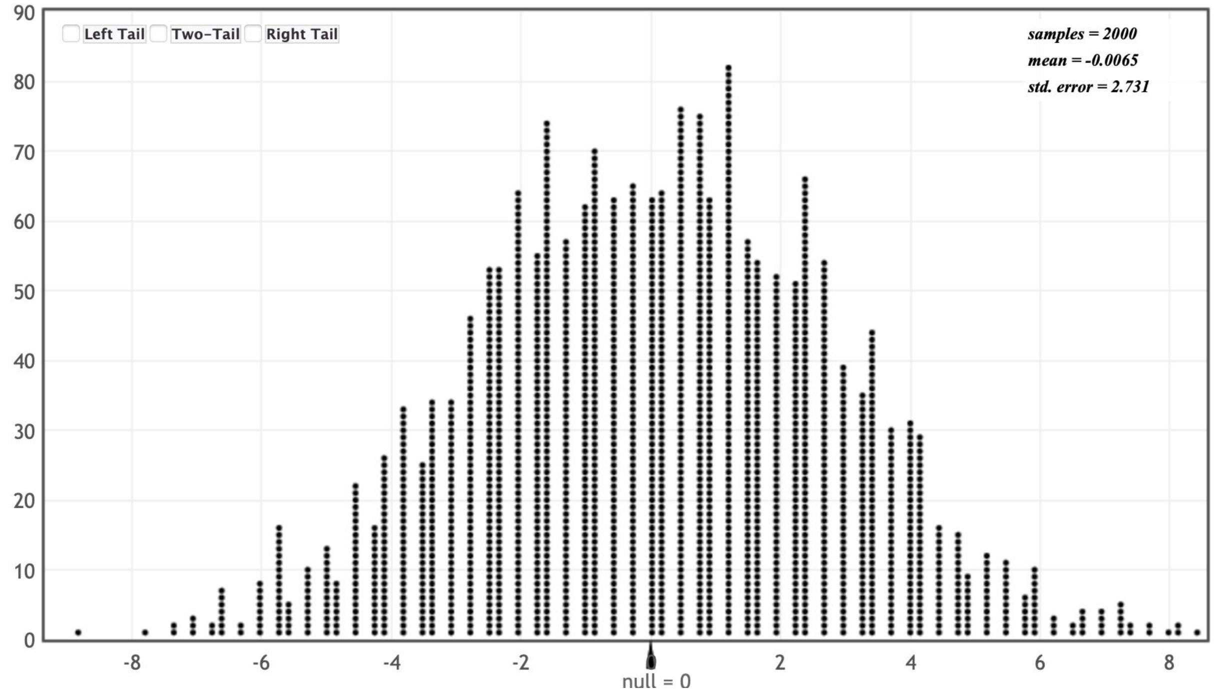
7. Write an alternative hypothesis for this situation: The difference in drinks did have an effect in the calcium excretion. Being in the diet cola group caused the woman to have more calcium excretion. Ha: Mc-Mw>0 or Mw-Mc<0

You should consult the null and alternative hypotheses we wrote for the two experiments we have studied this week to get an idea for how these should be worded! Write them in words and using symbols.

Go to StatKey, here.

Make sure to select the correct data set - go to the top left corner and click on the dropdown menu that says “Leniency and Smiles”. Instead, select “Cola and Calcium Excretion”.

8. Generate a dotplot for a randomization hypothesis. Paste a screenshot of your dotplot here.



9. What does each dot in the dotplot represent? Be specific and speak in context.

Every dot in the dotplot represents a the difference of the means in the simulation of woman drinking water or diet cola assuming the null is true

10. How many dots are in your dotplot?

2000

11. What are your thoughts about how the experimental difference of 6.875mg compares to what we see in the dotplot? Does 6.875mg seem unusual in comparison or not?

The 6.875 mg does seem unusual, the random dotplot that we have used shows that the mean for the data is4.38. We would then fail to reject the null hypothesis

12. Quantify how unusual the experimental result is by calculating a p-value. Remember, a p-value is the probability of getting a result as or more extreme than the experimental result under the assumption of the null hypothesis. To estimate this, count the number of dots that are greater than or equal to the experimental result and divide by the total number of dots. There are 8 so the p-value will be 8/2000= .0004

13. Write a paragraph for the researchers to explain your conclusion for this randomization test. You may borrow some of what you said above and you should include the p-value you calculated.

Assuming there is no difference in urine calcium between soda drinkers and water drinkers, the difference in the means in 6.875mg is extremely unusual. Out of the 2000 simulated experiments, only 8 results were more extreme than our mean. Which shows in the p-value of .0004. The probability of the result very small which means we can reject the null hypothesis. This means that the diet cola has a effect on calcium excretion.